# Haoyi Duan

Tel: (86) 19883161889 E-mail: scihaoyi2023@outlook.com
Website: https://haoyi-duan.github.io Github: https://github.com/haoyi-duan

## Education Background

**Zhejiang University, Hangzhou, China** **09/2019-06/2023**
B.Eng. Computer Science and Technology, **Chu Kochen Honors College**
**GPA**: **3.97**/4.00 (**91.38**/100) The 3rd year GPA **3.99**/4.00 (**93.17**/100)
**Ranking**: **1st** / 227
- ✧ **National Scholarship** awarded in 2021-2022 (TOP 1%)
- ✧ **First-Class Scholarship** awarded in 2019-2020, 2020-2021, and 2021-2022 (TOP 2%)

**National University of Singapore, Singapore** **05/2022-07/2022**
School of Computing Summer Workshop </>
- ✧ Took the course **Real-Time Graphics Rendering** (A+)
- ✧ Programmed a ray-tracing system and won the Third Prize (Ranking: 3/18)

## Publications & Patents

**Time as Prompt for Geography-aware Next Location Recommendation** </>
Yan Luo, **Haoyi Duan**, Ye Liu, Chung Fu-Lai, *WWW'2023 (Under Review)*

**Beyond Two-Tower Matching: Learning Sparse Retrievable Interaction Models for Recommendation** </>
Liangcai Su, **Haoyi Duan**, Jieming Zhu, Fan Yan, Xi Xiao, Zhou Zhao, Zhenhua Dong, Ruiming Tang, *WWW'2023 (Under Review)*

**MPI, OpenMP, and Taichi Parallel Programming Languages** </>
**Haoyi Duan**, *Electronic Components and Information Technology 2022*

**Lock-in Amplifier Measures Weak Sound Pressure Signals** </>
**Haoyi Duan**, *Computer Knowledge and Technology 2021*

**OpenGL-based "OpenCity" Game**, national patent no.:2022SR0805651, Copyright Protection Center of China

**UE4-based Self-developed FTPS "JYDGE: GENESIS" Game** (patent pending), Copyright Protection Center of China

## Internship

Intern Algorithm Engineer, **Huawei Noah's Ark Lab, China** **06/2022-08/2023**
- ✧ Work with Search & Recommendation Group
- ✧ Contribute to high recommendation and search accuracy with machine learning and NLP technologies in cross-modal domain

## Research Projects

**Time as Prompt for Next Location Recommendation** </> **08/2022-11/2022**
Supervisor: Prof. Chung Fu-Lai, The Hong Kong Polytechnic University
- ✧ Proposed a Temporal Prompt-based and Geography-aware (TPG) framework
- ✧ Designed Temporal prompt to incorporate temporal information of next location and then devised Shifted window to augment geographic data for addressing the hard boundary problem
- ✧ Improved the effectiveness and superiority of recommendation under various settings, with a creative function of interval prediction, i.e., predicting the location that a user wants to go to at a certain time while the most recent check-in behavioral data is masked

**Sparse Retrievable Interaction for Recommendation** </> **05/2022-11/2022**
Supervisor: Researcher. Jieming Zhu, Huawei Noah's Ark Lab
- ✧ Proposed an all-to-all interaction framework SparCode for recommendation to support a code-based sparse inverted indexing mechanism
- ✧ Improved the accuracy with enhanced feature interactions while keeping inference efficient
- ✧ Explored its feasibility in tasks such as cross-modal retrieval

**Pre-training Audio Representations** **03/2022-08/2022**
Supervisor: Prof. Zhou Zhao, Zhejiang University DCD Lab
- ✧ Improved the effectiveness of audio-related downstream tasks from massive amounts of unlabeled data through pre-training and high-quality audio representations
- ✧ Achieved state-of-the-art level in both cover recognition and sound scene classification tasks with a novel self-supervised audio representation learning framework, Audio-PAW, that contains three key components--Hierarchical Audio Cropping, Interaction Unit and Hierarchical contrastive

- ✧ Brought 14% accuracy improvements on music tagging and has been launched in Huawei's music recommendation services

**VQA Training Based on MindSpore** </> **02/2022-06/2022**

Supervisor: Prof. Siliang Tang, Zhejiang University DCD Lab

- ✧ Output the correct answer from the inputted question that contained an image and the content of the image in the form of natural language
- ✧ Applied a bi-channel image plus text model to obtain the probability of each answer to the question
- ✧ Used the Stack Attention model to filter the noise and obtain the highly answer-related region
- ✧ Realized a joint embedding of inputting an image and a question and a multi-label classifier based on a fixed set of candidate answers with the Top-down Attention model

**Attribute Classification of COVID-19-Related Tweets, Student Research Training Program** </> **04/2021-05/2022**

Supervisor: Prof. Siliang Tang, Zhejiang University DCD Lab

- ✧ Created a neural network for the automated identification of a COVID-19 tweet's binary attribute
- ✧ Data Augmentation by mutual translation on different language training datasets in the data preprocessing stage
- ✧ Used Bert, RoBERTa, and XLM-RoBERTa to build pre-training models
- ✧ Utilized LSTM+Attention, MultiHead Attention, the TextCNN model, and the adjustable Loss function in creating the classification models
- ✧ Trained 12 models that far surpass the best average F1-score of 89.7%, including RoBERTa-lstm-attn (91.38%), xlmRoBERTa-lstm-attn (91.09%), xlmRoBERTa-lstm-attn-biasedWeight (90.67%), xlmRoBERTa-multihead (90.49%), etc., optimized the training result by adopting voting mechanism and reached an ultimate best-vote of 93.54%

## Course Projects

**"OpenCity" Game Project (Supervisor: Prof. Hongzhi Wu)** </> **09/2021-01/2022**

- ✧ Developed the "OpenCity" inspired by "SimCity"
- ✧ Rendered images within the OpenGL framework

**Sokoban Game Software Programmed in x86 (Supervisor: Dr. Honghuan Bai)** </> **05/2021-07/2021**

- ✧ Converted C language of a puzzle video game genre to assembly language versions

**"Baba Is You" Game (Supervisor: Dr. Haifeng Liu, Dr. Qingsong Shi, and Prof. Wenzhi Chen)** </> **09/2020-01/2021**

- ✧ Programmed the replica game "Baba Is You" on FPGA based on Verilog
- ✧ Designed the single-cycle, multi-cycle, pipelining CPU with the functions such as Cache buffer and interrupt trigger

## Society Memberships & Extracurricular Activities

**Head of Saxophone Voice, Zhejiang University Marching Brass Band** **09/2019-06/2023**

- ✧ Organize and join the regular training and performances (as a player of Level 10 Certificate in Saxophone), including the National Day Flag Raising, the Student Festival Tour, Zhejiang University's 2019 New Year's Gala

**Player, Sports Games of Zhejiang University** **10/2020-05/2022**

- ✧ The 1st Place in Sanhao Cup Zhejiang University Rumba Dance Competition 05/2022
- ✧ The Final Four in the Basketball Tournament of Chu Kochen Honors College 05/2021
- ✧ The 8th Place in 400m hurdles in Zhejiang University Sports Meeting 10/2020

**Officer of Cultural and Creative Center, CKC Housing Committee** **09/2020-09/2021**

**Volunteer, Zhejiang University Volunteer Association** **09/2019-06/2021**

- ✧ Participated in over 120 hours of volunteer activities such as study room management, information session at High-School Alma Mater, organizing books at the children's branch of Hangzhou Library, and taking the referee role in the university sports meeting

**Senior Tutors Program of Chu Kochen Honors College, Zhejiang University** **08/2020-09/2020**

- ✧ Helped first-year students adapt to university life and study and won the Excellent Tutor Award in May 2021

## Languages & Skills

Languages: Chinese (native) English (proficient) **TOEFL: 108(S23) GRE: (V155+Q169+3.5)**

Programming: C/C++, Python, CUDA, mini SQL, Java, Verilog, x86, Shell

Tools/Software: Pytorch, OpenGL, MindSpore, MySQL, LATEX, Vivado, Adobe PS/Pr

## Prizes & Honors

Excellent Developer, awarded by Huawei on outstanding contribution to the open-source community 2022

Bronze Prize in the 8th "Intel Cup" Parallel Application Challenge 2020

Third Prize of the College Students Physics Innovation (Theoretical) Competition 2020